ARAVINDHAN S

Tiruvannamalai, Tamil Nadu | 6369942117 | aravindhanaravindhan737@gmail.com
linkedin.com/in/aravindhan-s-088041253 | leetcode.com/u/Aravindhan1812
Portfolio: aravindhan18sivaji.github.io

PROFESSIONAL SUMMARY

AI Systems Engineer with hands-on experience developing cost-effective, hybrid AI architectures using Python and FastAPI. Specialized in combining Text Intelligence via advanced Retrieval-Augmented Generation (RAG) with Numerical Intelligence through custom Machine Learning and Deep Learning models built from scratch. Proficient in leveraging Model Context Protocol (MCP) tools for seamless cross-platform integrations and agentic workflows. Strong focus on building deterministic backend systems and significantly reducing LLM inference and token costs through strategic text, audio, and semantic caching to deliver scalable, production-ready AI solutions.

TECHNICAL SKILLS

Programming & AI Development
Python | Object-Oriented Programming (OOP) | AsyncIO | FastAPI | REST APIs | End-to-End System Design | AI-Generated Code Cleanup & Refactoring | Compute & Resource Estimation for Process Allocation

Web Scraping, Data Extraction & Monitoring
Web Crawling | HTML Parsing | Google Search API | YouTube Transcript API | Unstructured Data Processing | LangSmith | LLM-as-a-Judge | AI Feedback Loops | Structured Logging | Argument Validation Evals

Machine Learning & Deep Learning
PyTorch | Scikit-Learn | Neural Networks | Machine Learning Algorithms | Deep Learning | Model Development from Scratch | Model Evaluation | Feature Engineering | Data Preprocessing | Computer Vision | YOLO | Vision Transformers (ViT) | Natural Language Processing (NLP) | Recommender Systems | Graph Neural Networks | GraphSAGE | KNN | Locality Sensitive Hashing (LSH)

Generative AI
Large Language Models (LLMs) | Transformers | Retrieval-Augmented Generation (RAG) | RAG Evaluation | Prompt Engineering | Agentic AI Systems | LangChain | LangGraph | CrewAI | Model Context Protocol (MCP) | Fine-Tuning | Vector Databases | FAISS | Chroma | General Vectorization | LLM Inferencing | vLLM | KV Caching | Paged Attention | Quantization | Batching

AI Infrastructure & Deployment
Docker | Celery | Redis | Redis Pub/Sub | Distributed Task Queues | Event-Driven Architecture | Asynchronous Inference | Git | GitHub Actions | Pytest | Prometheus | Input & Output Guardrails | Loop Engineering | Sandboxing | API Prefix Caching | Batch API Processing | Semantic RAG Caching

Data, Analytics & Databases
Data Cleaning | Data Preprocessing | Exploratory Data Analysis (EDA) | Missing Value Handling | Outlier Detection | Feature Scaling | Cross-Validation | PostgreSQL | MySQL | SQLAlchemy | Alembic

WORK EXPERIENCE

AI Systems Engineer

Vankaal

02/2026 - 08/2026

- Developed a real-time AI voice consultation system using Python, FastAPI, and Twilio for automated technical support.
- Integrated LLM-based communication capabilities with custom deterministic planning and business-logic components to maintain controlled AI decision-making.
- Built asynchronous backend workflows for handling real-time voice processing, conversation state, context tracking, and AI responses.
- Developed high-throughput WhatsApp automation workflows using the Meta Graph API and custom webhooks.
- Implemented backend state management and multi-tenant communication workflows for reliable processing of concurrent user requests.
- Integrated AI services into application workflows through APIs and asynchronous processing architectures.
- Intelligent Email Automation & Data Extraction: Architected an event-driven AI workflow that categorizes incoming emails, extracts structured information from PDFs and invoices, and generates context-aware responses using NLP pipelines.
- Built preprocessing workflows to transform unstructured documents and incoming data into structured information for downstream AI processing.
- Algorithmic Recommender Systems: Designed and implemented personalized recommendation systems using statistical machine learning, KNN, LSH, and graph-based similarity techniques.

- Developed recommendation algorithms from scratch using mathematical and machine learning fundamentals rather than relying entirely on high-level APIs.

Generative AI Developer

AI Futurix Innolabz

02/2025 - 02/2026

- Developed Python-based microservices for integrating and serving AI models through backend APIs.
- Built FastAPI services for AI-powered applications using asynchronous processing and REST-based API integration.
- Developed and integrated Generative AI solutions involving LLMs, RAG pipelines, agentic workflows, and AI infrastructure.
- Implemented data and message-processing workflows using AsyncIO, Redis, and asynchronous task architectures.
- Used Docker to package and maintain consistent AI application environments across development and deployment.
- Developed secure API endpoints using OAuth2 authentication and JWT-based authorization.
- Implemented automated testing using Pytest to validate AI application components and backend services.
- Monitored AI application and service performance using Prometheus and investigated application-level issues.
- Worked with AI models and backend systems to integrate machine learning capabilities into practical software applications.

KEY PROJECTS

Real Estate AI Agent & Multichannel Web-Scraping Bot (WhatsApp & Email)

Tech Stack: Python, Web Scraping, Google Search API, YouTube Transcript API, Webhooks, FastAPI, LLMs, Redis, PostgreSQL, LangSmith

- Architected a centralized, event-driven real estate AI assistant to autonomously handle complex customer inquiries across WhatsApp and email.
- Engineered dynamic web scraping and search pipelines using Google Search APIs to extract real-time location data (e.g., nearby schools and amenities) for specific plots.
- Integrated YouTube Transcript extraction and web crawling to parse video and web content, enabling the agent to read location details and provide credible YouTube video links to buyers.
- Processed diverse WhatsApp input modalities, including audio messages and PDFs, standardizing unstructured inputs into a unified NLP pipeline.
- Implemented shared semantic caching with Redis to reduce repeated LLM calls and optimized database handling for reliable state management across user sessions.
- Implemented structured logging and LangSmith monitoring, utilizing LLM-as-a-judge evaluations and automated feedback loops to validate agent tool arguments and improve response accuracy.

Real-Time IVR (Interactive Voice Response) Automation

Tech Stack: Python, FastAPI, Twilio Voice, WebSockets, Silero VAD, Groq Whisper Large-v3, Gemini 2.5 Flash, Streaming TTS, RAG, Redis, WhatsApp Business Platform, SMS

- Developed an automated telephone voice assistant using Twilio Voice and a FastAPI backend.
- Used bidirectional WebSocket media streaming to receive caller audio and return generated speech during live calls.
- Integrated Silero Voice Activity Detection (VAD) to identify speech activity in incoming audio and avoid sending silence unnecessarily for transcription.
- Integrated Groq Whisper Large-v3 for highly accurate speech-to-text transcription of incoming caller audio.
- Connected Google Gemini 2.5 Flash to generate responses from caller transcripts and relevant conversation context.
- Integrated a streaming Text-to-Speech (TTS) pipeline to convert generated responses into audio and send them back through the Twilio voice stream.
- Implemented barge-in handling so caller speech during assistant playback can trigger task cancellation and a Twilio clear event to remove queued audio.
- Used recent conversation turns in the active prompt and RAG retrieval to bring relevant historical conversation context into the response workflow.
- Implemented shared semantic caching with Redis to reuse responses for repeated queries, such as pricing, and reduce repeated LLM calls.
- Organized prompts to support provider-side prompt caching where available.
- Integrated Twilio SMS for inbound message handling, notifications, and follow-up communication.
- Integrated WhatsApp messaging through the Meta Graph API so the system can send relevant product images to a caller's WhatsApp during an ongoing phone call.

- Used WhatsApp image sharing to support product sales, product presentations, and visual clarification while the caller continues the voice conversation.
- Maintained system reliability with structured logging and LangSmith observability, leveraging LLM-as-a-judge evaluations to monitor argument passing and create continuous conversational feedback loops.

Voice-to-Voice AI Conversational System

Tech Stack: Python, Voice Processing, Conversational AI, Async Workflows, LangSmith

- Architected a voice chat application allowing users to interact directly with an AI assistant in real time.
- Optimized the processing pipeline to reduce response latency and support natural conversational turn-taking.
- Managed conversation state and context tracking to maintain relevant responses throughout the dialogue.
- Integrated comprehensive logging and LangSmith monitoring, utilizing LLM-as-a-judge to evaluate response accuracy and maintain robust AI feedback loops.

Graph-Based Recommendation Engine

Tech Stack: Python, Graph Neural Networks, Text Embeddings, Evals

- Designed a personalized recommendation system that matches users with relevant items based on node relationships.
- Integrated text embeddings to represent and compare unstructured data for recommendation.
- Built recommendation logic and mathematical aggregation components for matching and retrieval.
- Applied structured logging and LLM-as-a-judge evaluations to monitor retrieval quality and validate prediction feedback loops across the system.

EDUCATION

Bachelor of Technology / Bachelor's Degree in Artificial Intelligence and Data Science

Dr. N.G.P. Institute of Technology

10/2021 - 04/2025

GPA: 8.0 / 10

CORE STRENGTHS

- Machine Learning & Deep Learning
- AI Application Development
- Web Scraping & Data Extraction
- RAG System Development
- Distributed AI Workflows
- Algorithmic Problem Solving
- Generative AI & LLM Integration
- Model Development from Scratch
- AI / ML API Integration
- AI Backend Engineering
- Model Evaluation & Observability
- Mathematical Foundations of AI